

Chapter 3

Classical Probability, Counting and Combinatorics

Lecture 3 | Devore 9e: 2.2–2.3

Last time: set theory, experiments, events — and no probability anywhere. Today we assign numbers to events for the first time, in the oldest and simplest way.

3.1 Classical (Laplace) probability

Roll a fair die. $S = \{1, 2, 3, 4, 5, 6\}$. What is “the probability” here? We assign each outcome a number, and symmetry says they should all get the same one:

$$\mathbb{P}(1) = \mathbb{P}(2) = \cdots = \mathbb{P}(6) = \frac{1}{6}.$$

Definition 3.1 (Classical probability). Let $S = \{\omega_1, \dots, \omega_n\}$ be *finite* with all outcomes equally likely, $\mathbb{P}(\omega_1) = \cdots = \mathbb{P}(\omega_n) = \frac{1}{n}$. For an event $A \subset S$ containing m outcomes,

$$\mathbb{P}(A) = \frac{m}{n} = \frac{|A|}{|S|}$$

The triple $(S, \text{“events”}, \mathbb{P})$ — where “events” means all subsets of S — is our first probability model. Two things to notice:

- **Computing $\mathbb{P}$ is exactly counting.** There is no probability theory left in the formula; there is only m and n .
- **Counting can be very tricky.** That is the real content of this lecture, and of most exam problems in the first month.

3.2 Two fundamental principles of counting

Theorem 3.2 (Addition principle). *If a task can be done in m ways or in n ways, and the two groups of ways do not overlap, the total is $m + n$.*

Theorem 3.3 (Multiplication principle). *If a task is done in two steps, the first in m ways and the second in n ways, the total is $m \times n$. With k steps, multiply all k counts.*

Example 3.4 (Dining halls — addition). Three dining halls on campus offer 10, 8 and 6 entrées. If you will eat at exactly one of them, you have

$$10 + 8 + 6 = 24$$

choices. “Or” means add — and note this is only correct because no entrée is counted twice.

Example 3.5 (Passcode — multiplication). A 4-digit phone passcode is built by choosing each digit in turn, 10 ways each:

$$10 \times 10 \times 10 \times 10 = 10^4 = 10,000.$$

“And then” means multiply.

Remark 3.6. “Or” $\Rightarrow$ add; “and then” $\Rightarrow$ multiply. Most counting errors come from using the wrong one, or from steps that are not independent of each other (the count at step 2 must not depend on *which* choice was made at step 1 — only on how many there were).

3.3 Permutations and combinations

The organizing question: *you have n distinct objects and select k of them. How many ways?* The answer depends on two yes/no questions:

Does **order** matter?

Is the selection **with replacement**?

That is four cases. We do all four.

3.3.1 Case 1: order matters, without replacement

Build the selection in k steps:

step 1: n choices
 step 2: $n - 1$ choices
 $\vdots$
 step k : $n - k + 1 = n - (k - 1)$ choices

Definition 3.7 (Permutation). The number of ways to arrange k objects out of n distinct

objects, order mattering, without replacement, is

$$A_n^k := n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!},$$

read “ n permute k ”.

Taking $k = n$ gives $A_n^n = n!$: the number of ways to order all n objects.

3.3.2 Case 2: order does not matter, without replacement

Here is the trick that makes this case free. Every unordered selection of k objects can be ordered in $A_k^k = k!$ ways, so

$$\underbrace{A_n^k}_{\text{order matters}} = \underbrace{(\# \text{ order does not matter})}_{\text{what we want}} \times k!.$$

Definition 3.8 (Combination).

$$C_n^k = \binom{n}{k} := \frac{A_n^k}{k!} = \frac{n!}{k!(n-k)!},$$

read “ n choose k ”. Equivalently $A_n^k = \binom{n}{k} \times k!$.

Remark 3.9 (Two notations). I write C_n^k on the board and $\binom{n}{k}$ in typed material; they are the same number. Likewise A_n^k is often written $P(n, k)$ or nPk elsewhere.

Example 3.10. Take $n = 4$ objects $\{1, 2, 3, 4\}$ and $k = 2$. Ordered: $A_4^2 = 12$ pairs $(1, 2)$, $(2, 1)$, $(1, 3)$, $(3, 1)$, ... Unordered: $\binom{4}{2} = 6$ sets $\{1, 2\}$, $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$. And indeed $12 = 6 \times 2!$.

Example 3.11 (Candy — the relation made concrete). There are 10 different candies.

- *Pick 3 to eat for dessert.* You will eat all three, so the order does not matter: $\{\text{red, green, blue}\}$ is the same dessert as $\{\text{green, red, blue}\}$. That is $\binom{10}{3} = 120$ ways.
- *Pick 3 and hand them to Alice, Bob and Carol.* Now it matters who gets which. Each of the 120 selections can be handed out in $3! = 6$ ways, so

$$\binom{10}{3} \times 3! = 120 \times 6 = 720 = A_{10}^3.$$

The second bullet *is* the identity $A_n^k = \binom{n}{k} k!$: choosing an ordered list is the same as choosing the set and then ordering it.

3.3.3 Case 3: order matters, with replacement

Each of the k steps now has all n objects available again:

$$\underbrace{n \times n \times \cdots \times n}_k = n^k.$$

3.3.4 Case 4: order does not matter, with replacement

This one is genuinely tricky, and the standard trick is worth knowing.

Since order does not matter, all that is recorded is *how many times each object was selected*. Say object i was chosen x_i times. Then we need the number of integer solutions of

$$\sum_{i=1}^n x_i = k, \quad x_i \geq 0.$$

The balls-and-bars model. Draw the k selections as k balls in a row. To split them into n groups we need $n - 1$ bars:

$$\underbrace{\circ\circ}_{x_1} \mid \underbrace{\circ\circ}_{x_2} \mid \underbrace{\circ}_{x_3} \mid \underbrace{\circ\circ\circ}_{x_4}$$

There are $n + k - 1$ positions in total (balls plus bars), and choosing which $n - 1$ of them are bars determines everything:

$$\boxed{\binom{n+k-1}{n-1}}$$

3.3.5 The summary table

	without replacement	with replacement
order matters	$A_n^k = \frac{n!}{(n-k)!}$	n^k
order does not matter	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$	$\binom{n+k-1}{n-1}$

Remark 3.12. Memorize the table, but more importantly memorize the two *questions*. On an exam the hard part is never the formula — it is deciding which box you are in.

Example 3.13 (Shared birthdays). Each of k people has a birthday equally likely to be any of 365 days, independently of the others. What is the probability that at least two of them share a birthday?

A sample point is an ordered list of k birthdays, so $|S| = 365^k$ — order matters, with replacement. “All birthdays different” is order matters, *without* replacement: $A_{365}^k = 365 \cdot 364 \cdots (365 - k + 1)$ outcomes. Complementing,

$$\mathbb{P}\{\text{some shared birthday}\} = 1 - \frac{A_{365}^k}{365^k} = 1 - \prod_{i=0}^{k-1} \left(1 - \frac{i}{365}\right).$$

This is 0.476 for $k = 22$ and 0.507 for $k = 23$: with just 23 people a shared birthday is already more likely than not.

Remark 3.14 (Estimating it without a calculator). For small x , $1 - x \approx e^{-x}$, which turns the product into a single exponential:

$$\prod_{i=0}^{k-1} \left(1 - \frac{i}{365}\right) \approx \exp\left(-\sum_{i=0}^{k-1} \frac{i}{365}\right) = \exp\left(-\frac{k(k-1)}{730}\right).$$

Setting this equal to $\frac{1}{2}$ gives $k(k-1) \approx 730 \ln 2 \approx 506$, and indeed $23 \cdot 22 = 506$. The answer is small because what matters is the number of *pairs*: 23 people form $\binom{23}{2} = 253$ pairs, each matching with probability $1/365$, and $253/365 \approx 0.69 \approx \ln 2$.

3.4 Poker: answering Lecture 1's question

A five-card hand is dealt from a standard 52-card deck ($13 \text{ ranks} \times 4 \text{ suits}$). Order does not matter and there is no replacement, so we are in the lower-left box of the table:

$$|S| = \binom{52}{5} = 2,598,960.$$

Every hand is equally likely, so classical probability applies and each computation below is just “count the favorable hands”.

Royal flush — 10, J, Q, K, A all in one suit. There is exactly one such hand per suit, so 4 hands:

$$\mathbb{P} = \frac{4}{2,598,960} = 0.000154\%.$$

Straight flush — five consecutive ranks, all one suit. A straight is determined by its lowest card, which can be A, 2, 3, ..., 10 (with A playing low in A2345 and high in 10JQKA), so there are 10 possible runs per suit and $4 \times 10 = 40$ straight flushes. Excluding the 4 royal flushes, which are already ranked above:

$$\mathbb{P} = \frac{40 - 4}{2,598,960} = \frac{36}{2,598,960} = 0.001385\%.$$

Four of a kind — build it in two steps. Choose the rank that appears four times (13 ways); all four of its cards are then forced. Choose the remaining fifth card from the other 48 cards (48 ways):

$$\mathbb{P} = \frac{13 \times 48}{2,598,960} = \frac{624}{2,598,960} = 0.024010\%.$$

Full house — three of one rank plus two of another. Four steps, multiplied:

$$\underbrace{13}_{\text{rank of the triple}} \times \underbrace{\binom{4}{3}}_{\text{which 3 suits}} \times \underbrace{12}_{\text{rank of the pair}} \times \underbrace{\binom{4}{2}}_{\text{which 2 suits}} = 13 \times 4 \times 12 \times 6 = 3,744,$$

$$\mathbb{P} = \frac{3,744}{2,598,960} = 0.144058\%.$$

Note the 12, not 13: the pair must be a *different* rank from the triple.

Flush — all five cards one suit. Choose the suit (4) and then 5 of its 13 cards, $\binom{13}{5} = 1287$. That gives $4 \times 1287 = 5,148$, but this count includes the 40 straight flushes, which rank higher:

$$\mathbb{P} = \frac{5,148 - 40}{2,598,960} = \frac{5,108}{2,598,960} = 0.196540\%.$$

Straight — five consecutive ranks, suits arbitrary. Choose the run (10 ways), then a suit for each of the five cards independently ($4^5 = 1024$), giving 10,240; again remove the 40 straight flushes:

$$\mathbb{P} = \frac{10,240 - 40}{2,598,960} = \frac{10,200}{2,598,960} = 0.392465\%.$$

Hand	Count	Probability
Royal flush	4	0.000154%
Straight flush	36	0.001385%
Four of a kind	624	0.024010%
Full house	3,744	0.144058%
Flush	5,108	0.196540%
Straight	10,200	0.392465%

Remark 3.15. There is the answer to Question 1 of Lecture 1. The ranking is not a convention: each hand beats the one below it because it is strictly rarer. A flush outranks a straight because there are 5,108 flushes and 10,200 straights — a straight is almost exactly twice as likely.